

LCFSTE: Landslide Conditioning Factors and Swin Transformer Ensemble for Landslide Susceptibility Assessment

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Abstract—Landslide susceptibility assessment (LSA) holds crucial importance in guiding regional disaster prevention and reduction efforts. However, current deep learning (DL) models for LSA encounter challenges like insufficient landslide data samples and uneven distribution. In this article, we develop a new hybrid framework named landslide conditioning factors and swin transformer ensemble (LCFSTE), which integrates landslide conditioning factors (LCFs) and swin transformer (swin-T) for LSA. With this framework, we fully leverage the powerful nonlinear feature extraction capability of swin-T to extract abstract features from both landslides and LCFs. This approach ultimately enhances the precision and reliability of LSA. To assess the performance of our newly proposed framework, we selected Jiuzhaigou County, China, as our study area. First, a dataset for LSA was constructed using historical landslide data and 11 multisource LCFs. Then, these factors were screened through a multicollinearity test and factor importance analysis using variance inflation factor, tolerance, and information gain rate. Subsequently, the dataset was divided into three subsets: 60% for training, 20% for validation, and 20% for testing. Then, the LSA results were compared with

four DL models. Seven evaluation metrics (EMs) are chosen to quantitatively evaluate the performance of these five LSA models. The results demonstrated that, among these seven EMs, LCFSTE outperformed the others, achieving the highest score in six out of the seven considered EMs. This outcome highlights the promising applicability of LCFSTE in enhancing LSA accuracy.

Index Terms—Deep learning (DL), Jiuzhaigou County, landslide susceptibility assessment (LSA), swin transformer (swin-T).

I. INTRODUCTION

LANDSLIDE susceptibility is a measure of the likelihood of landslides occurring in a specific area based on local conditions, indicating the potential impact of slope movement on the landscape [1]. It serves as a predictive tool, estimating the areas where landslides are prone to happen. On August 8, 2017, the Jiuzhaigou County in China experienced a devastating earthquake with a magnitude of seven. This seismic event triggered a significant number of coseismic landslides and resulted in the emergence of numerous potential landslide hazards. Landslide susceptibility assessment (LSA) plays a pivotal role as the cornerstone of landslide disaster risk assessment. It provides valuable insights to decision-makers involved in disaster management and facility construction planning [2].

The significance of LSA extends to regional disaster prevention and mitigation efforts, actively contributing to the reduction of future landslide risks in the Jiuzhaigou area. Recent advances in LSA research can be divided into three areas: statistical methods (STMs), machine learning (ML) models, and deep learning (DL) models. STMs entail the inference of data relationships and offer advantages such as simplicity and strong interpretability. The earliest origins of LSA research can be traced back to methods like logistic regression [3], [4], [5], analytical hierarchy process [6], [7], information value [5], weight of evidence [8], and frequency ratio [9], [10], [11]. While STMs hold several advantages, they also come with some common limitations, including cumbersome procedural steps, lower computational efficiency, and reduced precision.

In recent years, many scholars have applied ML models to the field of LSA [12], [13]. In comparison to STMs, ML techniques prioritize computational efficiency and accuracy over interpretability. ML methods commonly employed in LSA encompass artificial neural networks [14], [15], support vector machines [16], [17], and random forests [18], [19]. Unlike

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statistical approaches, ML algorithms do not necessitate specialized hardware devices, such as GPUs, resulting in lower computational costs. However, before deploying a model into the network, feature engineering becomes imperative to reduce feature complexity and enhance the prominence of relevant features. This task, although crucial, can be challenging and resource-intensive.

DL is a subset of ML that employs deep neural networks to create more complex models, thereby achieving a deeper understanding of the underlying data. While it demands powerful hardware resources, DL eliminates the need for extensive feature engineering and excels when dealing with large datasets. DL can be categorized into five main types: convolutional neural networks (CNNs)-based [20], [21], [22], [23], [24], recurrent neural networks (RNNs)-based [25], [26], deep generative models [27], [28], [29], graph neural networks [30], and attention-based mechanisms [31], [32], [33]. Wang et al. [34] conducted the LSA of Jiuzhaigou County based on the transformer model and discovered that its accuracy surpassed that of the traditional ML model significantly. Zhou et al. [35] proposed a feature fusion transformer module built on the basis of the transformer framework foundation to fuse all landslide conditioning factors (LCFs) spatial information features. Yuan and Chen [36] introduced a spatiotemporal transformer aimed at enhancing the temporal probability prediction of landslides, thereby effectively improving the accuracy of landslide hazard assessment.

With the continuous improvement of computing power, DL has achieved great success in the computer vision field and has been widely used in LSA tasks. However, these DL models mainly focus on CNN-based and RNN-based models, which extract deep information by stacking convolutional layers, focusing only on local associations and failing to make effective use of global information [37], [38]. Compared with these models, transformer model does not contain convolution, pooling, or cyclic operations, which makes it have better parallelism and interpretability when processing sequence data. At the same time, the self-attention mechanism can capture dependencies between different positions in the input sequence, which allows the transformer to better handle global contextual information in visual tasks. Swin transformer (swin-T) stands as an enhancement over traditional transformer models, which has powerful nonlinear and multiscale feature extraction capability when compared to other DL models. Swin-T employs a local attention window strategy to effectively capture local information while processing large-scale images. This approach serves to mitigate computational complexity and enhance model efficiency. In addition, swin-T introduces a cross-layer connectivity mechanism, enabling information exchange across different levels within the model architecture. This enhances feature reusability and contributes to overall performance improvements.

Based on this, we propose a new framework that integrates LCFs with the swin-T. The main contributions of our work can be summarized as follows: 1) we introduce a hybrid framework named landslide conditioning factors and swin transformer ensemble (LCFSTE), which combines LCFs and swin-T for LSA

and 2) the proposed LCFSTE is applied to LSA in a landslide-prone area, improving the LSA accuracy of the framework.

The rest of this article is organized as follows. Section II comprehensively introduces the study areas and overviews the datasets employed in the experiments. Section III outlines the experimental steps undertaken to develop the new framework. Section IV analyzes the experimental results in detail, presenting both qualitative and quantitative findings. Finally, Section V concludes this article with some remarks.

II. STUDY AREA AND DATA

A. Study Area

The study area (see Fig. 1) is in a region of Jiuzhaigou County, situated in the northern Aba Tibetan and Qiang Autonomous Prefecture, Sichuan Province, China. The area lies within the latitude of $33^{\circ}5'59''$ – $33^{\circ}14'58''$ N and longitude of $103^{\circ}42'13''$ – $103^{\circ}51'53''$ E, covering an area of 250 km². The region features a river system, with the Baihe river running across the entire county and flowing into the Baishuihe river. It is a typical deep-incised valley located in the transition zone between the Qinghai-Tibet Plateau, Western Sichuan Plateau, mountains, and Sichuan Basin. Its elevation varies from 2641 to 4455 m, with a peak difference of 1814 m. As a result, this location presents various morphological processes and landforms, making it prone to the development of geo-hydrological hazards.

B. Landslide Inventory Map

In this article, the landslide inventory map was obtained through the visual interpretation from pre- and postevent Google Earth images (see Fig. 1). A total of 2171 landslides were identified, covering a combined area of 8.392 km². The largest landslide in terms of area was 0.5265 km², whereas the smallest measured 13.9581 m². Most of the landslides in the study area are relatively small and are concentrated in the vicinity of the epicenter, fault zones, and valley slopes.

C. Landslide Conditioning Factors

Due to the complexity of landslide development mechanisms and the differences between landslide types, there are no guidelines for selecting LCFs. In this study, based on previous studies and expert knowledge, 11 LCFs were selected: elevation, slope, terrain relief (TER), aspect, plan curvature (PLC), profile curvature (PRC), distance to faults (DTF), lithology, distance to rivers (DTR), modified normalized difference water index (MNDWI), and normalized difference vegetation index (NDVI). The first six factors were extracted from the digital elevation model with a spatial resolution of 30 m, which was downloaded from the geospatial data cloud.¹ The fault and lithology data were obtained by vectorization, based on the 1:200 000 scale geological map database of the National Geological Archive.² The river vector illustration was modified based on the 1:250 000 scale national basic geographic database, combined with

¹[Online]. Available at: <http://www.gscloud.cn/search>.

²[Online]. Available at: <http://www.ngac.org.cn/Map/List>.

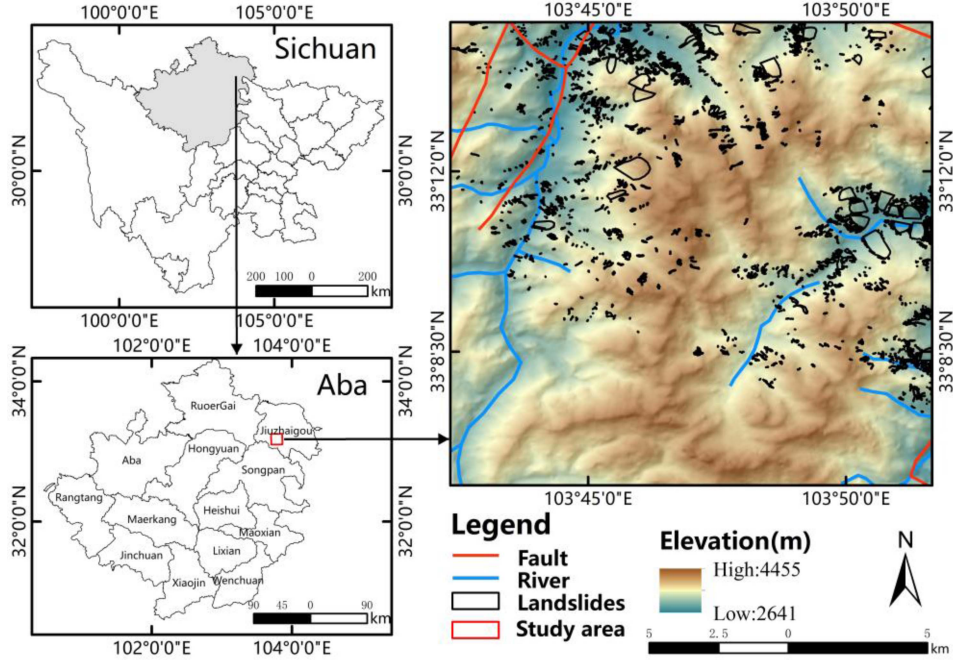


Fig. 1. Location of the study area.

TABLE I
DETAILED VALUES OF LCFs

LCFs	Values
Elevation (m)	[2641, 4455]
Slope (°)	[0, 73.48]
TER	[10, 483]
Aspect	Flat, North, Northeast, East, Southeast, South, Southwest, West, Northwest
PLC (°/100m)	[-4.80, 3.95]
PRC (°/100m)	[-5.76, 6.35]
DTF (m)	[0, 9762.18]
Lithology	Tertiary System, Permian System, Triassic System, Carboniferous System
DTR (m)	[0, 7230.25]
MNDWI	[-0.39, 0.56]
NDVI	[-0.13, 0.30]

pre-earthquake remote sensing image data. In addition, Landsat 8 OLI Image was used to obtain MNDWI and NDVI data. Table I shows the detailed information of the LCFs.

III. METHODOLOGY

The proposed LCFSTE framework encompasses three main parts (see Fig. 2). The first part is data processing, which includes data preprocessing, LCFs selection, and dataset creation for the LCFSTE. The second part of the framework ensembles the LCFs and swin-T algorithms. Finally, model evaluation is conducted in the third part of the framework. Seven evaluation metrics (EMs) are employed to compare the performance of our proposed LCFSTE framework to that of the other four DL methods.

A. Data Preprocessing

In our framework, data preprocessing involves two essential steps: resampling and image standardization. To ensure uniform

row and column numbers, resampling is conducted using the bilinear interpolation method, accounting for variations in data resolution [39]. Following resampling, all input data undergo standardization using the Z-score method, thereby achieving a consistent data distribution with a mean of 0 and a variance of 1. This standardization harmonizes data distributions across different dimensions and accelerates model convergence [40].

B. LCFs Selection

The process of selecting factors plays a pivotal role in determining the overall performance of the model. On one hand, a substantial presence of noise, characterized by factors with negligible or zero contribution rates in the dataset, can significantly distort the model's predictive outcomes, impeding the accurate assimilation of genuine landslide characteristics. On the other hand, robust interconnections among the factors may introduce data redundancy, ultimately diminishing the model's efficiency and exerting an influence on the allocation of feature weights throughout the learning process [41]. Consequently, in this study, we performed collinearity testing and importance analysis on the LCFs prior to the construction of the landslide sample datasets.

1) *Multicollinearity Test*: Multicollinearity is used to test whether there is nonperfect collinearity between LCFs. It can be judged by variance inflation factor (VIF) or tolerance (Tol). VIF is calculated as shown in (1). Tol is the reciprocal of the VIF

$$VIF = \frac{1}{1 - R_i^2} \quad (1)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

where y_i refers to the actual value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ refers to the average value.

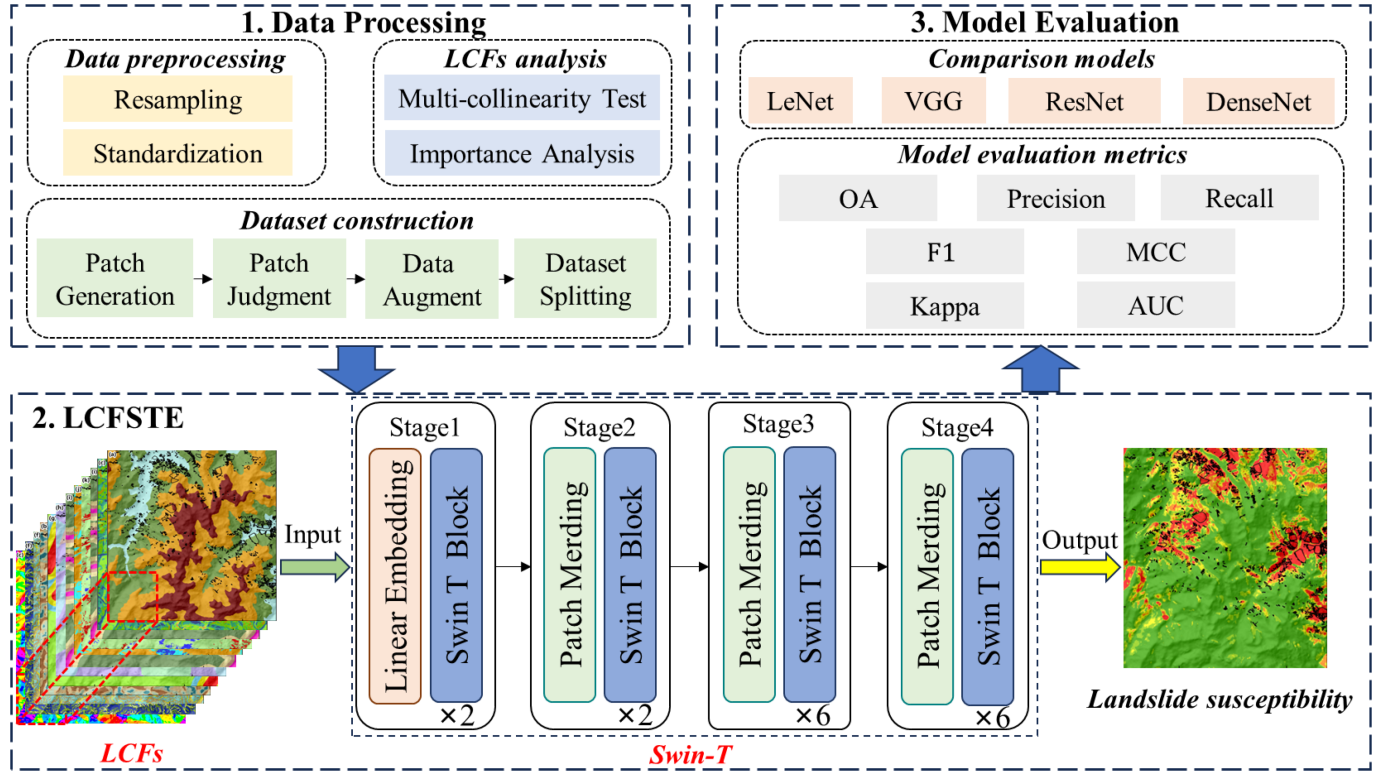


Fig. 2. Flowchart of LCFSTE.

In general, the calculation result of R_i^2 is between 0 and 1. The closer R_i^2 is to 1, the greater the number of independent variables that can be obtained through linear regression of other independent variables. Generally, a factor should be removed when its VIF exceeds 10 or 5.

2) *Importance Analysis:* To assess the influence of each LCF on the performance of the model, a comprehensive importance analysis of the 11 LCFs was conducted prior to model construction. In this study, we employed the information gain ratio (IGR) to determine the contribution of these LCFs to a landslide event [42]. The higher the value of a specific element, the more significant the corresponding LCF's contribution to the LSA model.

C. Dataset Construction

After extracting LCFs through data preprocessing and LCF analysis, it is essential to create a dataset before feeding it into the model. Given the spatial correlations of landslides at the pixel level, this study constructs the dataset using three-dimensional (3-D) sample blocks. To begin, the LCFs are assembled into a 3-D matrix (H, W, C), with H representing height, W representing width, and C indicating the number of channels (which corresponds to the number of LCFs). The subsequent steps involve patch generation and augmentation: first, it is assessed whether the majority of pixel labels within a sample block align with the central pixel label. If they match, the sample is retained; if there is a mismatch, it is considered an invalid sample. Following this evaluation, all landslide samples are selected, along with an equal number of nonlandslide samples. Subsequently, seven

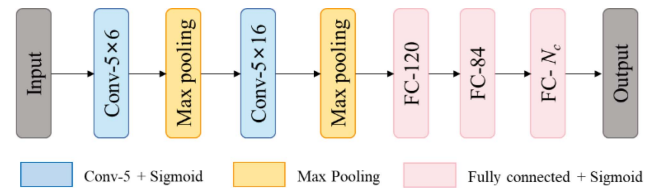


Fig. 3. Network structure of LeNet.

methods are employed for data augmentation, including 90°, 180°, and 270° rotations, left-right mirroring, and mirroring followed by 90°, 180°, and 270° rotations, effectively expanding the sample library to eight times its original size. Finally, the sample set is divided into training, validation, and test sets in a 6:2:2 ratio.

D. LeNet

LeNet is a relatively straightforward CNN, initially designed for recognizing handwritten text [43]. The classic LeNet-5 architecture comprises two 2-D convolutional layers for extracting features, each followed by a down-sampling layer, as well as a pooling layer for reducing feature dimensionality (see Fig. 3). Ultimately, it incorporates a classifier consisting of two fully connected (FC) layers and a Gaussian connected layer to produce classification results.

E. Visual Geometry Group Network (VGG)

The visual geometry group network (VGG) network employs a series of small consecutive convolutional kernels (3×3) and

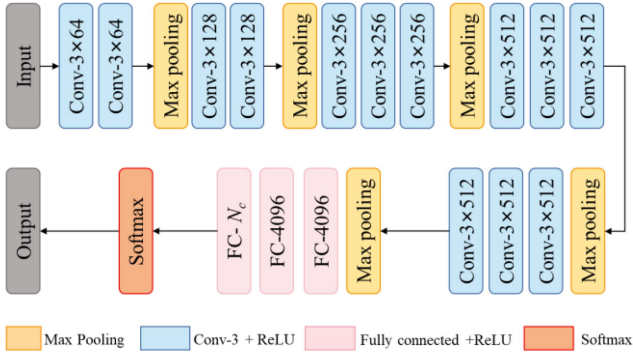


Fig. 4. Network structure of VGG.

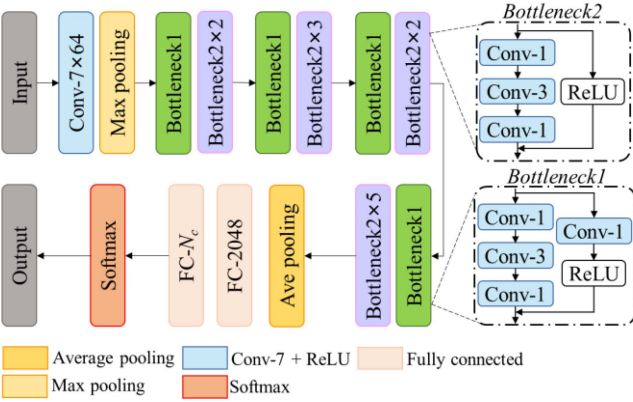


Fig. 5. Network structure of ResNet.

maximum pool layers (2×2) to build a deep neural network [44]. This architectural design allows for a total depth of either 16 layers or 19 layers. In our work, we employed the VGG16 model to map the landslide susceptibility, since it is one of the most widely used VGG networks. VGG16 consists of 16 hidden layers, including 13 convolutional layers (Conv-3) and pooling layers, followed by three FC layers. The Network structure of VGG is shown in Fig. 4.

F. Residual Neural Network (ResNet)

Residual neural network (ResNet) effectively addresses the challenges of gradient vanishing and exploding in the training of deep neural networks by introducing skip connections within the network architecture, resulting in significant enhancements in model performance [23]. At the heart of ResNet lies the Residual Block, comprised of two convolutional layers. Notably, the output of the second convolutional layer is combined with the input, creating a crucial junction that enables the residual block to glean and learn residual information. Finally, there is a global average pooling layer and a FC layer for classification.

The specific network employed in this study is ResNet50, distinguished by its inclusion of 50 convolutional and FC layers. This substantial depth empowers the network to capture and comprehend progressively intricate and abstract features. The ResNet network architecture is shown in Fig. 5.

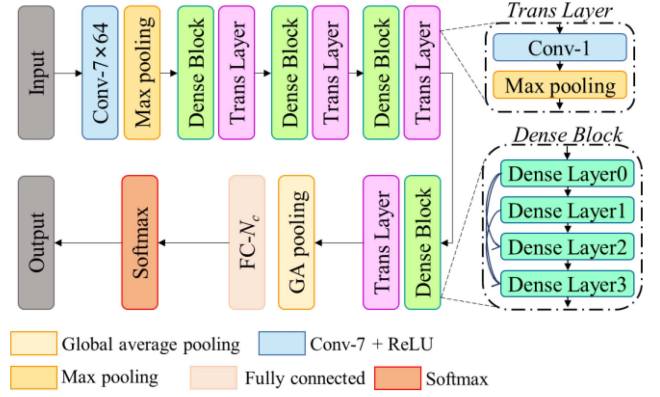


Fig. 6. Network structure of DenseNet.

G. Dense Convolutional Neural Network (DenseNet)

Dense convolutional neural network (DenseNet's) defining feature lies in the interconnections between every layer of dense layers. These dense connections within the Dense Block not only mitigate the vanishing gradient problem but also bolster feature propagation [45]. Furthermore, DenseNet incorporates a structure known as the dense block-transition to maintain consistent feature map sizes at each layer. Transition encompasses transition layers, comprised of a 1×1 convolution layer and a max-pooling layer. In pursuit of enhanced computational efficiency, the DenseNet-BC variant was conceived, with "B" signifying "Bottleneck." This variant comprises only two convolution layers, each with a preceding 1×1 convolution step to replace the conventional dense layer. The DenseNet architecture employed in this study is elucidated in Fig. 6, where the quantity of dense blocks in Stages 1–4 is specified as (6, 12, 24, 16).

H. LCFSTE

By integrating LCFs and swin-T, the LCFSTE model not only enhances the accuracy of LSAs but also significantly improves computational efficiency. The second part of Fig. 2 illustrates the specific structure of LCFSTE. In the context of LCFSTE, the swin-T network structure plays a pivotal role in feature extraction. The key breakthrough of swin-T lies in its innovative hierarchical grouping attention mechanism, which empowers LCFSTE to efficiently handle large-scale, high-resolution images, and capture global landslide features while demanding fewer computational resources.

When dealing with a patch block composed of LCFs, LCFSTE initiates the process by dividing the patch block into nonoverlapping image blocks through patch partition. Subsequently, LCFSTE is organized into four stages, with each stage comprising patch merging and multiple swin-T blocks. The specific architecture of these blocks is visually depicted in Fig. 7. It includes essential components like layer norm, multilayer perceptron, window multihead self attention (W-MSA), and shifted window multihead self attention (SW-MSA). Both W-MSA and SW-MSA compute the MSA within the window. SW-MSA, on the other hand, ensures connectivity among nonoverlapping windows and recalculates self-attention within the window of

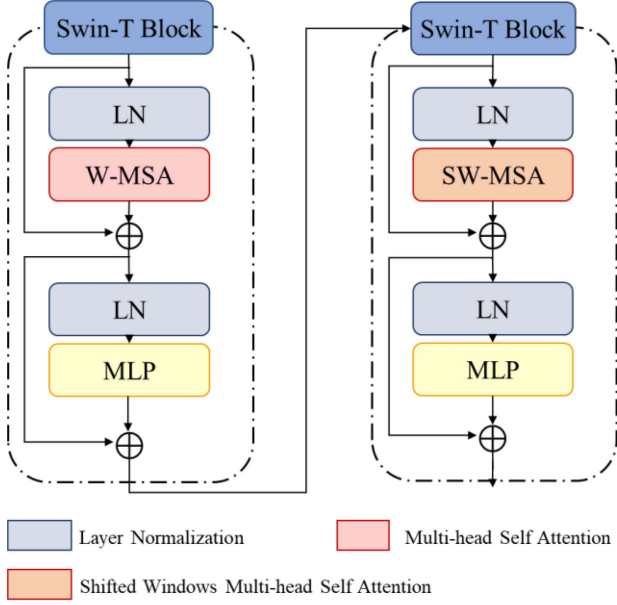


Fig. 7. Network structure of two connected swin-T blocks.

TABLE II
COMPUTER HARDWARE INFORMATION

Item	Parameter
CPU	Intel Xeon(R) Silver 4210R CPU@2.4GHZ×40
GPU	GeForce RTX 3090/PCIe/SSE2
Memory	128GB
Hard disk	8TB

the shifted feature map. These two swin-T modules are interconverted to facilitate information exchange between neighboring blocks.

I. Experimental Environment and Hyperparameter Optimization

Table II presents the computer hardware employed in the experiments conducted in this study. The software was developed within the Python 3.6.2 environment and built on the PyCharm 2021 and Microsoft Visual Studio 2015 software platform. The experiments were executed on the Linux-Ubuntu 18.04.05 LTS operating system. Furthermore, PyTorch was installed as the primary artificial intelligence development framework, with Keras and TensorFlow serving as supplementary tools.

Optimizing hyperparameters represents a pivotal phase in model debugging, as the right combination can yield maximum model performance. In this research, we employed the cross-entropy function as a loss function. The stochastic gradient descent algorithm was chosen as the optimizer as it not only converges rapidly but also displays strong performance across various accuracy metrics. The initial learning rate is set to 0.00007 and the batch size is set to 32 for each learning iteration. The network was trained for 100 epochs, and the loss value on the validation set was monitored during training.

J. Model EMs

In this article, seven EMs were selected to assess the performance of the LSA model, which were overall accuracy (OA), Precision (Pr), Recall (Re), F1 score (F1), Matthews correlation coefficient (MCC), Kappa coefficient (Kappa), and area under curve (AUC). These seven EMs are defined as follows.

1) *Overall Accuracy*: It measures the proportion of correct predictions made by a model over the total predictions. The calculation of OA is shown in the following equation:

$$OA = \frac{TP + TN}{TP + FP + TN + FN} \quad (3)$$

where true positive (TP) represents correctly predicted landslide samples and true negative (TN) stands for correctly predicted nonlandslide samples. A false positive (FP) occurs when the actual value is a nonlandslide, but the model incorrectly predicts the sample as a landslide. A false negative (FN) arises when the actual value is a landslide, but the model incorrectly predicts the sample as a nonlandslide.

2) *Precision*: Pr is the ratio of correctly predicted landslide observations to the total predicted landslide sample. Pr is calculated by the following equation:

$$Pr = \frac{TP}{TP + FP} \quad (4)$$

3) *Recall*: Re is the ratio of correctly predicted landslide observations to the actual landslide samples as shown in the following equation:

$$Re = \frac{TP}{TP + FN} \quad (5)$$

4) *F1 Score*: F1 is the harmonic mean of Pr and Re. It is a single metric that combines both Pr and Re into one value. It is calculated as follows:

$$F1 = 2 * \frac{Pr * Re}{Pr + Re} \quad (6)$$

5) *Matthews Correlation Coefficient*: MCC takes into account TP, TN, FP, and FN and ranges from -1 to 1. An MCC of 1 represents a perfect prediction, 0 is equivalent to random prediction, and -1 indicates total disagreement between prediction and observation. MCC can be obtained through the following equation:

$$MCC =$$

$$\frac{TL \times TN - FL \times FN}{\sqrt{(TL + FP) \times (TL + FN) \times (TN + FL) \times (TN + FN)}} \quad (7)$$

6) *Kappa Coefficient*: Kappa measures the agreement between observed and expected classifications while considering the possibility of the agreement occurring by chance. The equation of Kappa is shown as follows:

$$Kappa = \frac{OA - Pe}{1 - Pe} \quad (8)$$

$$Pe = \frac{(TP + FN)(TP + FP) + (FP + TN)(FN + TN)}{(TP + FP + TN + FN)^2} \quad (9)$$

TABLE III
RESULTS OF FACTOR MULTICOLLINEARITY TEST

LCFs	VIF	Tol
Elevation	2.316	0.432
Slope	2.478	0.404
TER	2.541	0.393
Aspect	1.214	0.824
PLC	1.197	0.835
PRC	1.224	0.817
DTF	1.438	0.695
Lithology	1.103	0.907
DTR	1.252	0.799
MNDWI	1.605	0.623
NDVI	2.19	0.457

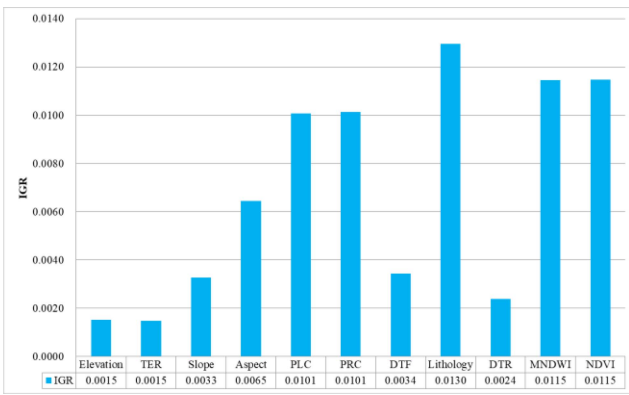


Fig. 8. Importance ranking distribution of each LCFs.

7) *Area Under Curve*: AUC measures the entire 2-D area underneath the entire receiver operating characteristic (ROC) curve. It is used to evaluate the performance of binary classification models and represents the model's ability to distinguish between positive and negative classes. AUCs close to 1 indicate high accuracy, whereas closer to 0.5 indicate low accuracy.

IV. RESULTS AND ANALYSIS

A. Analysis of LCFs

1) *Multicollinearity Test of LCFs*: The factor collinearity indexes of the study were calculated. The results in Table III show the Tol and VIF values of the study area. It can be seen that the VIF values of all LCFs are less than 5, which indicates that there is no need to remove any LCF.

2) *Importance analysis of LCFs*: The illustration in Fig. 8 displays the distribution of importance rankings for each LCF in the study area. Observably, the lithology factor holds the highest IGR value (0.0130), indicating its paramount contribution to the model. Next in rank are the MNDWI and NDVI factors, both at 0.0115, followed by the planar curvature and PRC factors. The elevation and TER factors exhibit the lowest IGR values (both at 0.0015), indicating their comparatively minimal contribution to the modeling process.

TABLE IV
EMs OF ALL LSA MODELS (THE BOLDDED AND UNDERLINED VALUE MEANS THE HIGHEST EMs OF THE ROW)

Model	OA	Pr	Re	F1	MCC	Kappa	AUC
LeNet	0.8429	0.7955	0.9232	0.8546	0.6949	0.6859	0.9238
VGG	0.8429	0.8144	0.8883	0.8497	0.6887	0.6859	0.8681
ResNet	0.8551	0.8083	<u>0.9311</u>	0.8654	0.7186	0.7103	0.9292
DenseNet	0.8626	0.8274	0.9162	0.8696	0.7293	0.7251	0.9382
LCFSTE	<u>0.8739</u>	<u>0.8475</u>	0.9119	<u>0.8785</u>	<u>0.7500</u>	<u>0.7478</u>	<u>0.9444</u>

B. Comparison of LSA Results

The landslide susceptibility maps are obtained by using the five LSA models. Then the natural breakpoint method [45], [46], [47] was used to divide the susceptibility category into five levels: very high (VH), high (HI), moderate (MO), low (LO), and very low (VL) (see Fig. 9). Based on the classification results of the natural breakpoint method, the classification thresholds for landslide susceptibility values obtained by LCFSTE are 0.10, 0.31, 0.56, and 0.81, respectively.

The area of VH zone is concentrated in and around the landslide sites, with HI and MO susceptibility category radiating outward. The LO and VL susceptibility areas cover the largest extent, with some scattered small landslides. However, it can be seen that the grain size of the susceptible regions in ResNet is significantly coarser compared to the other four LSA models. This suggests that the transition of the region is not smooth. The VH areas of large landslides are connected together, with the finest grain size observed in LCFSTE. In addition, the LSA results of LeNet seem not reasonable, as the central areas of some large landslides are categorized as MO even LO susceptible areas rather than VH areas. In addition, the LSA results of LeNet seem not reasonable, as the central areas of some large landslides are categorized as HI susceptible areas rather than VI areas. Therefore, visually, LCFSTE is the most reliable among the five LSA models.

Specifically with regard to the distribution of LSA results from the LCFSTE model, from a lithologic perspective, it can be found that HI and VH zones are mostly distributed within the carboniferous system, which is consistent with the fact that carboniferous system are usually susceptible to erosion. In terms of topography, HI and VH zones are mostly located in areas of undulating terrain and high slopes. This illustrates the importance of topographic elements for LSA.

C. Model Performance Evaluation

Assessing the metrics of OA, Pr, Re, F1, MCC, and Kappa (see Table IV), the LCFSTE model showcases superior values across most indicators. Notably, the ResNet model exhibits the highest Re value. The OA of the LCFSTE model stands at approximately 87.39%, with DenseNet following at 86.26%, ResNet at 85.51%, and the LeNet and VGG models ranking the lowest at 84.29%.

In terms of Re, the ResNet model attains the highest value at 0.9311, followed by LeNet at 0.9232, DenseNet at 0.9162, LCFSTE at 0.9119, and the least favorable value of 0.8883

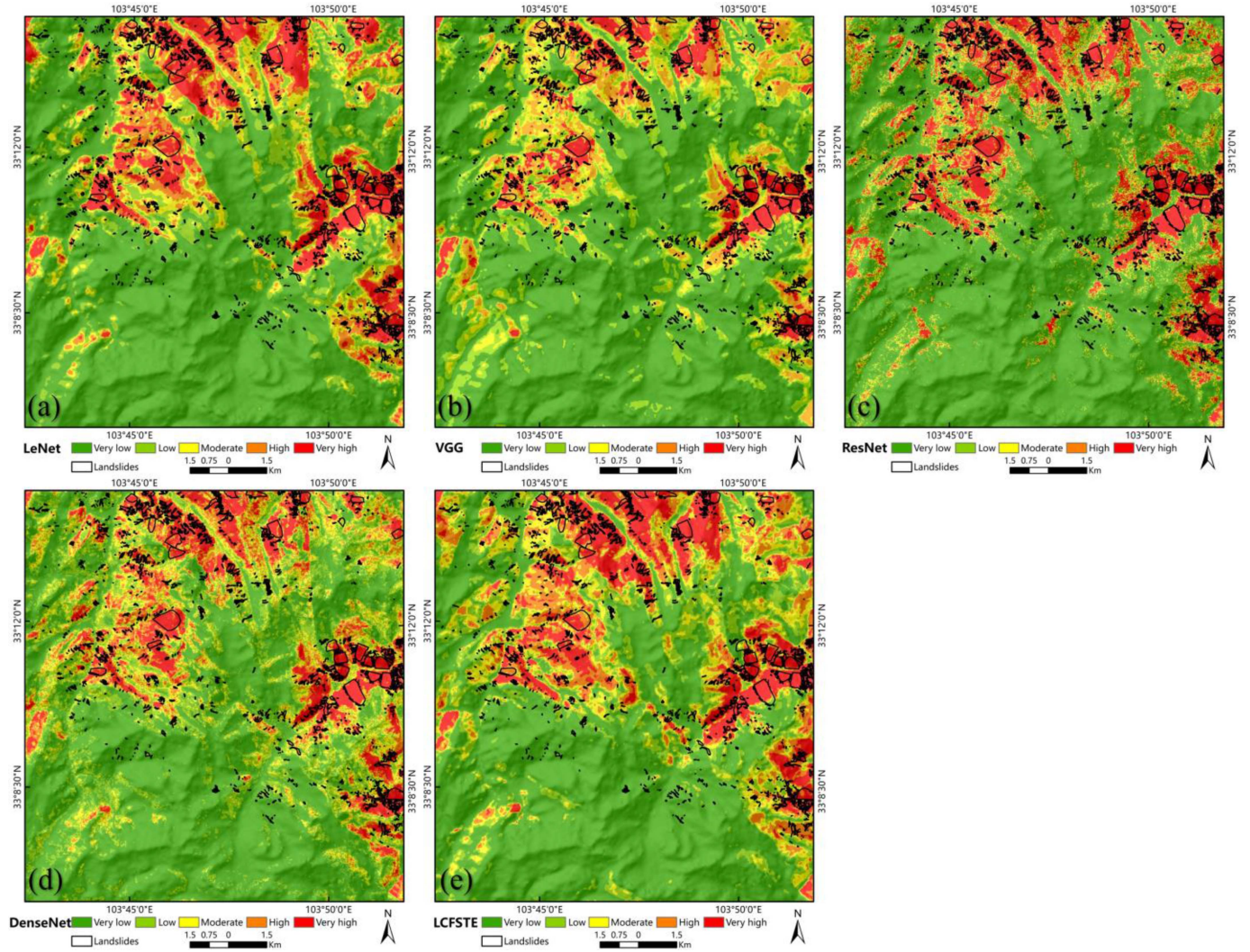


Fig. 9. Landslide susceptibility maps obtained by: (a) LeNet, (b) VGG, (c) ResNet, (d) DenseNet, and (e) LCFSTE.

from the VGG model. In addition, other indicators such as Pr, Kappa, F1, and MCC follow a similar trend as OA across the models, affirming that the LCFSTE model demonstrates superior predictive capabilities. Hence, it stands out as the most promising model for LSA task.

The ROC curves displaying the performance of five LSA models using the testing dataset are presented in Fig. 10. The LCFSTE model leads with the highest AUC value of 0.9444, followed by DenseNet at 0.9382, ResNet at 0.9292, LeNet at 0.9238, and the lowest AUC is attributed to VGG, registering 0.8681.

Based on the above-mentioned findings, it is clear that the LCFSTE model achieves the overall best performance among these five LSA models, since it resulted in the six highest accuracies out of the seven EMs.

V. DISCUSSIONS

In this article, the performance of the LCFSTE framework represents a remarkable advance over the other comparable models and models proposed by other scholars. For example,

Kikuchi et al. [48] reported an AUC value of 0.911 for their CNN model, whereas our LCFSTE model achieved an AUC value of 0.94, which shows that the optimization of the model mechanism can improve the prediction accuracy. Chen and Song [49] explore the potential applications of CNN hybrid ensemble metaheuristic optimization algorithms, namely beluga whale optimization and coati optimization algorithm, for landslide susceptibility mapping. The AUC values of the two optimized models are 0.906 and 0.919 respectively, which are lower than the LCFSTE model that has not been optimized in this article, reflecting the excellent performance of the LCFSTE model.

While the LCFSTE framework we introduced demonstrated robust performance in LSA, it faces significant challenges within this domain. First, the selection of LCFs may introduce uncertainties in the LSA due to variations in the importance of each LCF among the five methods (see Fig. 9). Lithology emerges with the highest importance index, followed by NDVI, MNDWI, PLC, PRC, aspect, DTF, slope, and DTR. Elevation and TER, on the other hand, register the lowest importance indices. This implies that in the study area, lithology significantly influences LSA, as different rock types have varying impacts on landslide

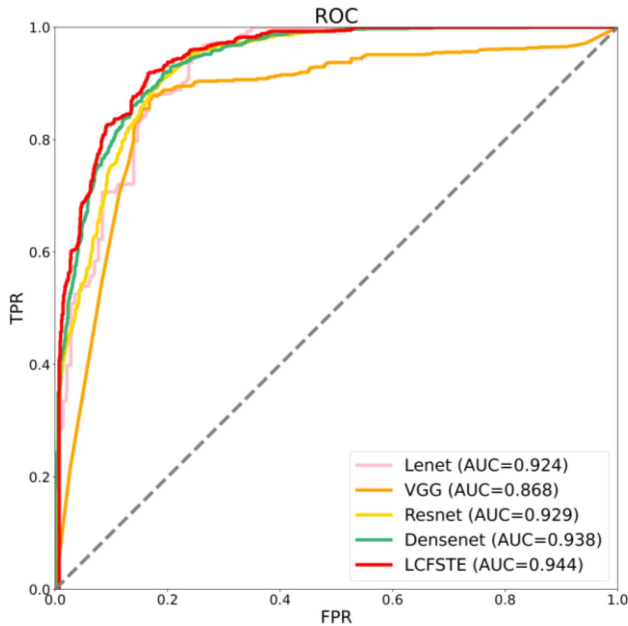


Fig. 10. ROC curves and AUC values on the testing dataset of the five LSA models.

occurrences. Despite efforts to calculate the importance of each LCF, discrepancies persist, introducing errors and diminishing LSA accuracy [50], [51], [52]. Second, our work did not classify different types of landslides when applying the five LSA methods, potentially introducing uncertainty. Prior studies have shown that landslide susceptibility can be more accurately assessed by considering individual types based on their distinct spatial incidences [4], [53]. Third, the absence of complete and coherent landslide inventory data in the area introduces further uncertainties. This deficiency may lead to unidentified new landslides in the study area, which might not be accounted for in existing landslide inventory datasets.

VI. CONCLUSION

In this article, we proposed a new LCFSTE hybrid framework, which ensembles LCFs and swin-T for LSA. We selected Jiuzhaigou County as our study area and constructed a dataset that included historical landslide data and eleven LCFs. We then conducted a multicollinearity test and importance analysis to screen the LCFs. Subsequently, the dataset was divided into three subsets for training, validation, and testing. To quantitatively assess the performance of the LCFSTE framework, we employed seven EMs and compared its results with four other DL models. The LSA results show that the spatial distribution of historical landslides and the VH and H zones obtained from the LCFSTE framework are in general agreement. Statistically, the LCFSTE framework outperforms the other four models in six out of the seven EMs. These findings underscore the LCFSTE framework's exceptional ability to extract abstract features from both landslides and LCFs. In summary, our study demonstrates the effectiveness of the LCFSTE framework in improving the accuracy of LSA.

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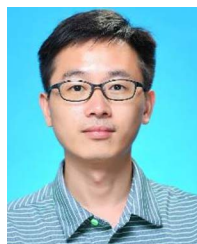
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